

Table 15. Computational-cost analysis of HyperPotter under three controlled sweeps on a server equipped with an AMD EPYC 7543 32-Core Processor and a single NVIDIA RTX A6000 GPU. **Default configuration:** cluster setting (16, 16, 20, 16, 20, 16), 4 s input, and $\text{max_iter} = 5$. **The default setting in each sweep is highlighted in bold for easy comparison.** Latency is reported as per-sample latency with batch size = 1; throughput and peak memory are reported with batch size = 128. For the cluster sweep, the six values in each configuration correspond to (S , T , ST11, ST12, ST21, ST22).

Experiment	Setting	Latency (ms) ↓	Throughput (utt/s) ↑	Peak Mem. (GiB) ↓	Mean Iter.	Mean Nodes
Cluster-count sweep (4 s input, $\text{max_iter} = 5$)						
K	0.25 N (11, 12, 14, 7, 14, 7)	60.27	38.52	13.76	5.0	42.8
K	0.50 N (21, 33, 27, 13, 27, 13)	76.98	23.84	12.20	5.0	42.8
K	0.75 N (32, 50, 41, 20, 41, 20)	94.65	16.67	13.41	5.0	42.8
K	HyperPotter (16, 16, 20, 16, 20, 16)	73.06	25.81	14.63	5.0	42.8
Node-count sweep (default cluster setting, $\text{max_iter} = 5$)						
N	1 s input	61.62	34.40	4.36	4.10	23.0
N	2 s input	62.03	34.99	7.49	5.00	29.5
N	4 s input	69.34	27.20	14.97	5.00	42.8
N	6 s input	74.85	23.75	22.46	5.00	55.7
N	8 s input	81.37	20.97	29.94	5.00	69.5
FCM iteration sweep (4 s input, default cluster setting)						
T	$\text{max_iter} = 1$	58.51	27.48	13.76	1.0	42.8
T	$\text{max_iter} = 3$	66.73	27.43	12.20	3.0	42.8
T	$\text{max_iter} = 5$	69.30	27.34	13.41	5.0	42.8
T	$\text{max_iter} = 7$	76.53	27.16	14.63	7.0	42.8
T	$\text{max_iter} = 10$	86.32	26.79	15.85	10.0	42.8

Table 16. Sensitivity of model performance to maximum number of FCM iterations T (default: $T = 5$), evaluated using EER (%).

	ITW	21LA	21DF	ASVspoof5	FoR
T=5 (default)	5.72	3.94	1.78	16.04	3.89
$T = 3$	5.79	3.19	2.25	19.95	5.21
$T = 7$	5.87	3.97	2.20	20.06	4.59

Table 17. Sensitivity of model performance to random initialization seeds across datasets, reported in EER (%). Results are reported under three random seeds for both the baseline and HyperPotter, and the mean $\pm$ std over seeds is shown in the last block. Lower EER indicates better performance, and the default seed is highlighted in bold.

Seed	Model	ITW	21LA	21DF	ASVspoof5	FoR
1234 (default)	Ours	5.72	3.94	1.78	16.04	3.89
	Baseline	7.58	2.48	4.08	13.38	4.24
1111	Ours	6.54	3.06	2.35	17.27	5.96
	Baseline	8.43	1.82	4.28	13.57	12.10
4321	Ours	6.40	3.79	1.91	14.92	4.13
	Baseline	8.08	3.10	3.75	14.18	13.16
Mean $\pm$ Std	Ours	6.22 $\pm$ 0.44	3.60 $\pm$ 0.47	2.01 $\pm$ 0.30	16.08 $\pm$ 1.18	4.66 $\pm$ 1.13
	Baseline	8.03 $\pm$ 0.43	2.47 $\pm$ 0.64	4.04 $\pm$ 0.27	13.71 $\pm$ 0.42	9.83 $\pm$ 4.87